

Primitive van der Waerden Numbers

A Standalone Statement and Elementary Proof of Problem 5

May 2026

Abstract

Problem 5 asks whether every finite coloring of a sufficiently long interval must contain a monochromatic arithmetic progression whose first term and common difference are coprime. We prove this by an elementary reduction to the classical van der Waerden theorem. More precisely, if $W(r, k)$ is the ordinary van der Waerden number, then

$$W(r, k) \leq W^*(r, k) \leq 1 + (W(r, k) - 1) \prod_{p \leq \lfloor (W(r, k) - 1) / (k - 1) \rfloor} p.$$

This proves in particular that $W^*(r, k) < \infty$ for all $r \geq 1$ and $k \geq 2$.

1 Statement of the problem

For a positive integer N , write

$$[N] = \{1, 2, \dots, N\}.$$

A k -term arithmetic progression is a sequence

$$a, a + d, a + 2d, \dots, a + (k - 1)d,$$

where a, d are positive integers and $d \geq 1$. We call it *primitive* if

$$\gcd(a, d) = 1.$$

Problem 5 (Primitive van der Waerden numbers). Let $r \geq 1$ and $k \geq 2$. Define $W^*(r, k)$ to be the least positive integer N , if such an integer exists, such that every r -coloring of $[N]$ contains a monochromatic primitive k -term arithmetic progression.

Does $W^*(r, k)$ exist? More quantitatively, how does it compare with the ordinary van der Waerden number $W(r, k)$?

Here an r -coloring of $[N]$ means a function

$$\chi : [N] \rightarrow \{1, 2, \dots, r\}.$$

A progression is monochromatic if all its terms have the same value under χ .

Recall that the ordinary van der Waerden number $W(r, k)$ is the least positive integer M such that every r -coloring of $[M]$ contains a monochromatic k -term arithmetic progression, with no coprimality condition. The classical van der Waerden theorem says that $W(r, k)$ exists for all $r \geq 1$ and $k \geq 1$.

2 Main theorem

For an integer $H \geq 1$, put

$$P_H = \prod_{p \leq H} p,$$

where the product is over primes $p \leq H$. If there is no prime at most H , the product is interpreted as 1.

Theorem 1. *Let $r \geq 1$ and $k \geq 2$, and let $W = W(r, k)$. Define*

$$H = \left\lfloor \frac{W-1}{k-1} \right\rfloor, \quad P_H = \prod_{p \leq H} p.$$

Then

$$W(r, k) \leq W^*(r, k) \leq 1 + (W(r, k) - 1)P_H.$$

In particular, $W^(r, k)$ exists and is finite.*

Proof. We prove the lower and upper bounds separately.

Lower bound. Let $N < W(r, k)$. By the definition of $W(r, k)$, there is an r -coloring of $[N]$ with no monochromatic k -term arithmetic progression at all. Such a coloring certainly has no monochromatic primitive k -term arithmetic progression, because every primitive arithmetic progression is, in particular, an arithmetic progression. Therefore no $N < W(r, k)$ can have the defining property of $W^*(r, k)$. Hence

$$W^*(r, k) \geq W(r, k),$$

provided $W^*(r, k)$ exists. The upper bound below proves existence.

Upper bound. Let

$$N = 1 + (W - 1)P_H.$$

We show that every r -coloring of $[N]$ contains a monochromatic primitive k -term arithmetic progression.

Take an arbitrary r -coloring

$$\chi : [N] \rightarrow \{1, 2, \dots, r\}.$$

Consider the W integers

$$1, 1 + P_H, 1 + 2P_H, \dots, 1 + (W - 1)P_H.$$

All of them lie in $[N]$. They form an affine copy of the interval $\{0, 1, \dots, W - 1\}$. Define an induced coloring

$$\psi : \{0, 1, \dots, W - 1\} \rightarrow \{1, 2, \dots, r\}$$

by

$$\psi(t) = \chi(1 + P_H t).$$

Since $\{0, 1, \dots, W - 1\}$ is just a translate of $[W]$, the definition of $W = W(r, k)$ implies that there exist integers $t_0 \geq 0$ and $h \geq 1$ such that

$$t_0, t_0 + h, t_0 + 2h, \dots, t_0 + (k - 1)h$$

all lie in $\{0, 1, \dots, W - 1\}$ and all have the same ψ -color.

Because the last term of this index progression is at most $W - 1$, we have

$$t_0 + (k - 1)h \leq W - 1.$$

Since $t_0 \geq 0$, it follows that

$$(k - 1)h \leq W - 1,$$

and therefore

$$h \leq \left\lfloor \frac{W - 1}{k - 1} \right\rfloor = H.$$

Now map this monochromatic progression of indices back into $[N]$. We obtain the k integers

$$1 + P_H t_0, \quad 1 + P_H(t_0 + h), \quad 1 + P_H(t_0 + 2h), \quad \dots, \quad 1 + P_H(t_0 + (k - 1)h).$$

They are all the same χ -color, by the definition of ψ . They also form an arithmetic progression. Its first term is

$$a = 1 + P_H t_0,$$

and its common difference is

$$d = P_H h.$$

It remains only to prove that this progression is primitive, namely that $\gcd(a, d) = 1$.

Let q be any prime divisor of d . Since

$$d = P_H h,$$

Euclid's lemma gives $q \mid P_H$ or $q \mid h$. If $q \mid P_H$, then q is one of the primes appearing in the product P_H . If instead $q \mid h$, then $q \leq h \leq H$, so again q appears in the product P_H . Thus in every case

$$q \mid P_H.$$

But then, since $a = 1 + P_H t_0$, we have

$$a \equiv 1 \pmod{q}.$$

Therefore $q \nmid a$.

We have shown that no prime divisor of d divides a . Hence no prime divides both a and d , and so

$$\gcd(a, d) = 1.$$

Thus the monochromatic progression constructed above is primitive. Since the original coloring χ was arbitrary, every r -coloring of $[N]$ contains a monochromatic primitive k -term arithmetic progression. Therefore

$$W^*(r, k) \leq N = 1 + (W - 1)P_H.$$

Combining this with the lower bound proves the theorem. $\square$

3 Immediate consequences

Corollary 1. *For every $r \geq 1$ and $k \geq 2$,*

$$W^*(r, k) < \infty.$$

Thus the qualitative existence part of Problem 5 follows from the ordinary van der Waerden theorem alone.

Proof. The ordinary van der Waerden theorem gives $W(r, k) < \infty$. Theorem 1 then gives an explicit finite upper bound for $W^*(r, k)$. $\square$

Corollary 2. *For every $k \geq 2$,*

$$W^*(1, k) = k.$$

Proof. With one color, the interval $[k]$ contains the primitive progression

$$1, 2, \dots, k,$$

whose first term is 1 and common difference is 1. Hence $W^*(1, k) \leq k$. On the other hand, no interval $[N]$ with $N < k$ can contain any k distinct positive integers, and therefore cannot contain a k -term arithmetic progression. Hence $W^*(1, k) \geq k$. $\square$

Corollary 3 (The case $k = 2$). *Let p_r denote the r th prime. For every $r \geq 1$,*

$$W^*(r, 2) = p_r.$$

Proof. A primitive two-term arithmetic progression is a pair

$$a, a + d$$

with $\gcd(a, d) = 1$. Since

$$\gcd(a, a + d) = \gcd(a, d),$$

primitive two-term progressions are exactly pairs of distinct coprime integers.

Consider the graph G_N whose vertex set is $[N]$ and in which two distinct vertices are adjacent precisely when they are coprime. An r -coloring of $[N]$ with no monochromatic primitive two-term progression is exactly a proper r -coloring of G_N . Thus every r -coloring of $[N]$ contains a monochromatic primitive two-term progression exactly when

$$\chi(G_N) > r,$$

where $\chi(G_N)$ is the chromatic number of G_N .

We now compute $\chi(G_N)$. The set

$$\{1\} \cup \{p : p \leq N, p \text{ prime}\}$$

is a clique in G_N , because 1 is coprime to every integer and distinct primes are pairwise coprime. Therefore

$$\chi(G_N) \geq \pi(N) + 1,$$

where $\pi(N)$ is the number of primes at most N .

Conversely, color the vertex 1 with its own color. For every $n > 1$, choose one prime divisor of n , for instance its least prime divisor, and color n by that prime. If two integers $m, n > 1$ receive

the same prime color p , then $p \mid m$ and $p \mid n$, so $\gcd(m, n) \geq p > 1$. Hence they are not adjacent in G_N . The vertex 1 is alone in its color class. This gives a proper coloring of G_N using exactly $\pi(N) + 1$ colors, one color for 1 and one color for each prime at most N . Thus

$$\chi(G_N) = \pi(N) + 1.$$

Therefore every r -coloring of $[N]$ forces a monochromatic primitive two-term progression exactly when

$$\pi(N) + 1 > r,$$

or equivalently when $\pi(N) \geq r$. The least positive integer N with at least r primes not exceeding N is the r th prime p_r . Hence

$$W^*(r, 2) = p_r.$$

□

4 A simple additional lower bound for $k \geq 3$

The lower bound $W^*(r, k) \geq W(r, k)$ is automatic. There is also a small arithmetic lower bound that shows the coprimality condition has some independent force.

Proposition 1. *For $r \geq 1$ and $k \geq 3$,*

$$W^*(r, k) \geq p_{r+1},$$

where p_{r+1} is the $(r + 1)$ st prime. Consequently,

$$W^*(r, k) \geq \max\{W(r, k), p_{r+1}\}$$

for $k \geq 3$.

Proof. We construct an r -coloring of

$$[p_{r+1} - 1]$$

with no monochromatic primitive k -term progression.

Use the first r primes as color labels:

$$p_1, p_2, \dots, p_r.$$

For every integer n with $2 \leq n \leq p_{r+1} - 1$, choose a prime divisor of n and color n by that prime. This is always possible using only the first r primes: every prime divisor of such an n is strictly smaller than p_{r+1} , and hence is among $p_1, \dots, p_r$. Color 1 arbitrarily, say with color p_1 .

Suppose there is a monochromatic k -term progression

$$a, a + d, \dots, a + (k - 1)d$$

under this coloring, with $k \geq 3$.

First suppose $a > 1$. If the common color is p_i , then every term of the progression is divisible by p_i . In particular,

$$p_i \mid a \quad \text{and} \quad p_i \mid a + d.$$

Subtracting gives $p_i \mid d$. Hence p_i divides both a and d , so

$$\gcd(a, d) > 1.$$

Thus the progression is not primitive.

It remains to consider the case where the progression contains 1. Since the progression consists of positive integers and has positive common difference, if it contains 1 then its first term must be $a = 1$. The common color must then be the color assigned to 1, which we chose to be p_1 . Because $k \geq 3$, the progression contains the two later terms

$$1 + d \quad \text{and} \quad 1 + 2d.$$

Both have color p_1 , so both are divisible by p_1 . Hence

$$1 + d \equiv 0 \pmod{p_1} \quad \text{and} \quad 1 + 2d \equiv 0 \pmod{p_1}.$$

Subtracting the first congruence from the second gives

$$d \equiv 0 \pmod{p_1}.$$

Substituting this into $1 + d \equiv 0 \pmod{p_1}$ gives

$$1 \equiv 0 \pmod{p_1},$$

which is impossible.

Therefore this coloring contains no monochromatic primitive k -term progression. Hence $p_{r+1} - 1$ does not have the defining property of $W^*(r, k)$, and so

$$W^*(r, k) \geq p_{r+1}.$$

□

5 Example

The ordinary van der Waerden number $W(2, 3)$ is 9. Therefore

$$H = \left\lfloor \frac{9-1}{3-1} \right\rfloor = 4, \quad P_H = 2 \cdot 3 = 6.$$

Theorem 1 gives

$$9 \leq W^*(2, 3) \leq 1 + (9-1) \cdot 6 = 49.$$

The numerical upper bound is not expected to be sharp; its significance is that it is finite, explicit, and obtained by a direct elementary reduction to the ordinary van der Waerden theorem.

6 Conclusion

Problem 5 has an elementary qualitative solution: the primitive van der Waerden numbers exist. In fact, the proof gives the explicit comparison

$$W(r, k) \leq W^*(r, k) \leq 1 + (W(r, k) - 1) \prod_{p \leq \lfloor (W(r, k) - 1) / (k - 1) \rfloor} p.$$

The construction works by looking only at the congruence class $1 \pmod{P_H}$, where P_H contains every possible prime divisor of the common difference produced by the ordinary van der Waerden theorem. This forces the first term to be congruent to 1 modulo every prime that could divide the common difference, and therefore forces the resulting progression to be primitive.